Assignment 1: Understanding JDBC Components

Task 1: What is JDBC? Explain its purpose.

Answer:

JDBC (Java Database Connectivity) is a Java API that allows Java applications to connect and interact with databases.

The primary purpose of JDBC is to establish a connection between a Java application and a database, execute SQL queries, retrieve results, and update data.

Using JDBC, developers can perform database operations such as:

- Connecting to a database
- Executing SQL queries
- Inserting records
- Updating records
- Deleting records
- Retrieving data
- Managing transactions

Advantages of JDBC

- Platform independent
- Supports multiple databases
- Easy database connectivity
- Secure using PreparedStatement
- Supports transaction management

Task 2: Explain the role of DriverManager, Connection, Statement, and ResultSet.

1. DriverManager

Role:

DriverManager manages JDBC drivers and establishes a connection between the Java application and the database.

Example

```
Connection con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/student","root","root1234");
```

---

2. Connection

Role:

Connection represents an active connection between the Java application and the database.

It is used to:

- Execute SQL queries
- Create Statement objects
- Commit/Rollback transactions

Example

```
Connection con = DriverManager.getConnection(url,user,password);
```

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3. Statement

Role:

Statement is used to execute SQL queries.

Types:

- Statement
- PreparedStatement
- CallableStatement

Example

```
Statement stmt = con.createStatement();

ResultSet rs = stmt.executeQuery("SELECT * FROM student");
```

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4. ResultSet

Role:

ResultSet stores the data returned from a SELECT query.

It allows the program to read records one by one.

Example

```
while(rs.next())
{
  System.out.println(rs.getString("name"));
}
```

Task 3: List at least three JDBC drivers and their uses.

1. MySQL JDBC Driver

Driver Name

com.mysql.cj.jdbc.Driver

Use

Used to connect Java applications with MySQL databases.

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2. Oracle JDBC Driver

Driver Name

oracle.jdbc.OracleDriver

Use

Used for connecting Java applications to Oracle Database.

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3. PostgreSQL JDBC Driver

Driver Name

org.postgresql.Driver

Use

Used for connecting Java applications to PostgreSQL databases.

Assignment 2

Case Study: Successful Implementation of Multi-Tier Architecture

Case Study: Amazon E-commerce Platform

Architecture

Amazon mainly follows a **Three-Tier Architecture**.

1. Presentation Layer

This is the user interface.

Technologies:

- HTML
- CSS
- JavaScript
- React
- Mobile Apps (Android/iOS)

Functions:

- Product search
- Login
- Registration
- Shopping cart
- Checkout

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2. Business Logic Layer

This layer processes user requests and business rules.

Technologies:

- Java
- Spring Boot
- REST APIs
- Microservices

Functions:

- User authentication
- Product management
- Order processing
- Payment processing
- Inventory management

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3. Database Layer

Stores all application data.

Technologies:

- MySQL
- PostgreSQL
- Amazon DynamoDB
- Amazon Aurora

Stores:

- Customer details
- Products
- Orders
- Payments
- Delivery information
